Step 1: Software Development Projects

Project Title:
Local Service Problem Diagnosis and Matching System

4. Nonfunctional Requirements

NFR-01 – Performance: The system can display provider search results within 2 seconds for 95% of requests.

NFR-02 – Response Time: The system can predict a service category within 3 seconds for 95% of problem descriptions.

NFR-03 – Security: The system requires login to access protected user, provider, and administrator functions.

NFR-04 – Privacy: Personal details are accessible only to authorized users.

NFR-05 – Usability: A new user can submit a service request within 3 minutes after logging in.

NFR-06 – Reliability: The system can save at least 99% of valid service requests during normal operation.

NFR-07 – Classification Accuracy: The NLP model targets at least 80% accuracy on a separate test dataset.

NFR-08 – Compatibility: The system works with the latest versions of Chrome, Edge, and Firefox.

NFR-09 – Maintainability: The application uses at least four separate code modules for request handling, provider matching, database operations, and text classification.

NFR-10 – Data Integrity: Each saved request contains a unique ID, a valid service category, and a valid status.

NFR-11 – Availability: The system targets 99% availability during scheduled operating hours.

NFR-12 – Data Backup: The system creates a database backup at least once every 24 hours.